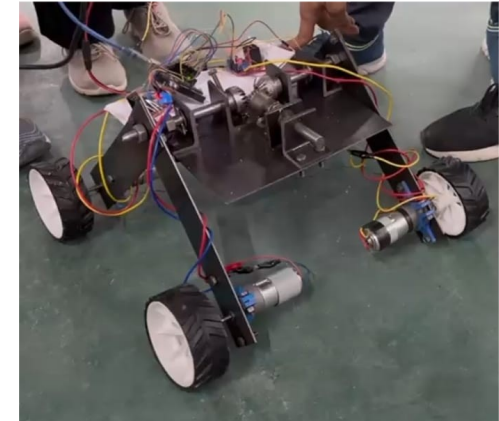




SPRINGLESS CAR SUSPENSION SYSTEM USING BEVEL GEARS

INSTRUCTOR- Prof. Virkeshwar Kumar

GUIDE – Mr. Rahul



GROUP 15

AKUL AGARWAL(230108)(C-4) KARAN KUMAR(230533)(C-4) SOHAM GIRISH PANCHAL(231014)(C-4) MEHER NARULA(230648)(C-5) SHRAVAN(230983)(C-5)



ANANYA KARTICK IYER (230133)(C-6)



KESHAV MEENA(230555)(C-6)



VEDANT NEEKHRA(231131)(C-6)



RONIT KUMAR(230875)



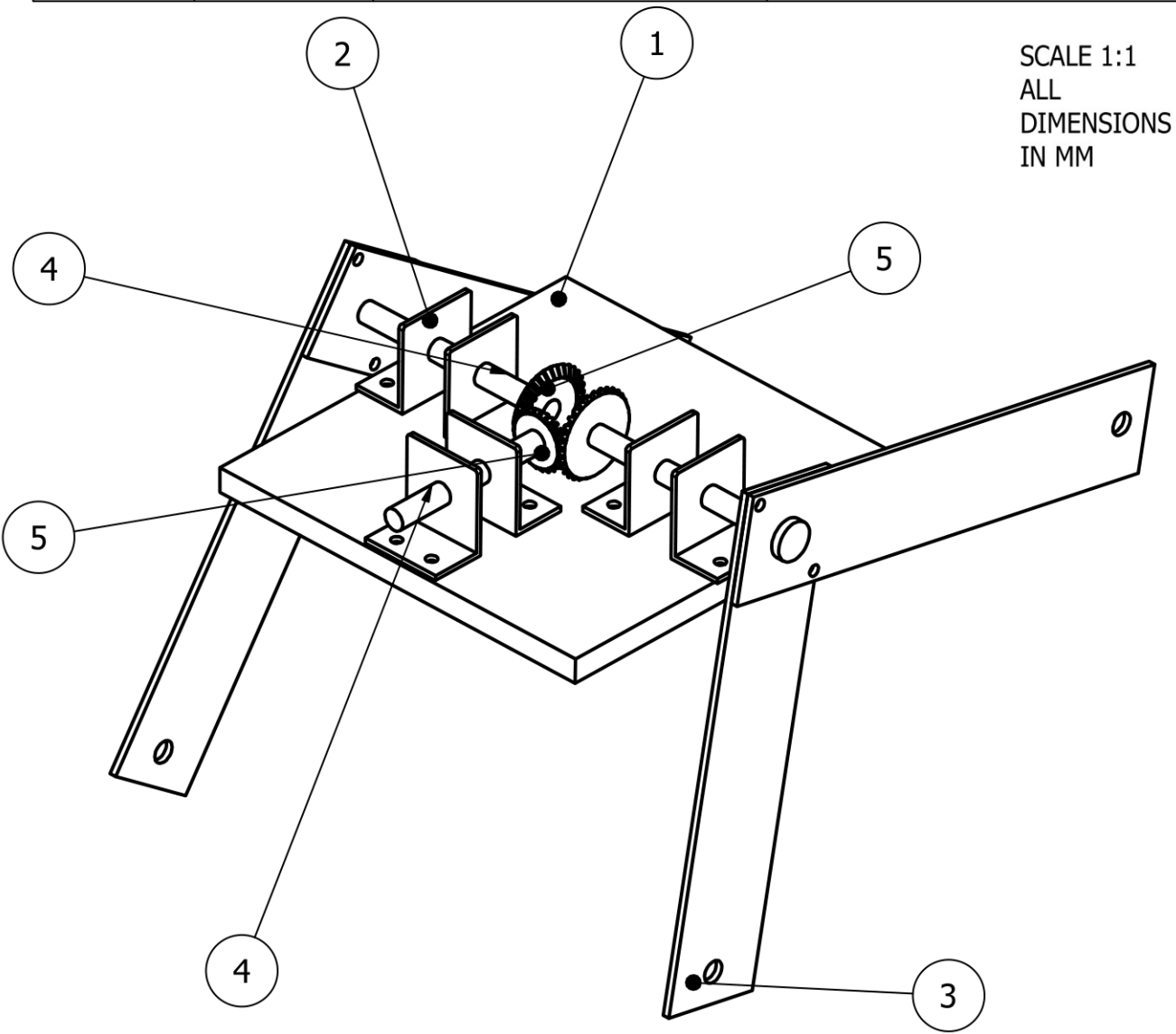
Project Overview

The objective of this project is to design and build a springless suspension system that absorbs shocks through bevel gears and keeps the whole vehicle on ground and not let any wheel get stuck in air. In total, 17 parts were manufactured, including bevel gears(3), L-supports(6), a base plate, legs(4), and shafts(3). The estimated project cost is Rs 5866.89. Future improvements could include making the system remote-controlled using a battery and Bluetooth, as well as enhancing the durability of the gear materials to extend the system's lifespan.

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PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	Base_Plate_Project	Dimension = 254x254 mm ² Thickness = 15mm
2	6	L-support	Dimension = 53x50x10 mm ³ Thickness = 3mm
3	4	Legs	Length = 300 mm Hole diameter = 6.50 mm
4	3	Shaft	Diameter = 12.7 mm Length = 110 mm
5	3	Bevel Gear	Module = 1.5 No. of teeth = 20 Pitch Diameter = 30 mm



Designed by Group 15	Checked by	Approved by	Date	Date 01-09-2024	
			Assemblydraft_isometric		
			Edition	Sheet 1 / 1	

5 Design Calculations

5.1 Motor Selection

For our rover, selecting the right motor was a critical part of the design process due to its substantial weight, which demanded motors capable of providing high torque. After assessing various options, we concluded that high-torque motors were essential to ensure the rover could handle the load effectively, even on uneven terrains. High torque was prioritized over speed (RPM). Thus, we settled on **High Torque 30 RPM 12V DC motor**.

5.2 Motor Driver Selection

To select an appropriate motor driver, we first measured the current draw for each pair of motors under different conditions. We found that each pair drew around **0.7 A without load** and **1.4-1.6 A with load**. Keeping both current requirements and cost in mind, we selected **two L298N motor drivers**. The L298N driver, with its **2A current rating**, provided enough margin to handle the loaded current draw safely while keeping the cost within budget.

5.3 Bevel Gear Calculations

1. Shaft angle, $\Sigma = 90^\circ$
2. Module, $m = 1.5$
3. Number of teeth, $z_1 = 20$ and $z_2 = 20$
4. Pitch diameter, $d = zm = 20 \times 1.5 = 30$ mm
5. Pitch cone angle,

$$\delta_1 = \tan^{-1} \left(\frac{\sin \Sigma}{\frac{z_1}{z_2} + \cos \Sigma} \right)$$

$$\delta_1 = \tan^{-1} \left(\frac{\sin 90^\circ}{\frac{20}{20} + \cos 90^\circ} \right)$$

$$\delta_1 = 45^\circ \quad \text{and} \quad \delta_2 = \Sigma - \delta_1 = 45^\circ$$

6. Cone distance, $R_e = \frac{d_2}{2 \sin \delta_2}$

$$R_e = \frac{30}{2 \sin 45^\circ}$$

$$R_e = 21.21 \text{ mm}$$

7. Face width, $b \leq \frac{R_e}{3}$ or $10m$

$$b \leq 7.07 \text{ mm}$$

8. Addendum, $h_a = m = 1.5$ mm

9. Dedendum, $h_f = 1.25 \times m = 1.25 \times 1.5 = 1.875$ mm

10. Dedendum angle, $\theta_f = \tan^{-1} \left(\frac{h_f}{R_e} \right)$

$$\theta_f = \tan^{-1} \left(\frac{1.875}{21.21} \right)$$

$$\theta_f = 5.05^\circ$$

11. Addendum angle, $\theta_a = \tan^{-1} \left(\frac{h_a}{R_e} \right)$

$$\theta_a = \tan^{-1} \left(\frac{1.5}{21.21} \right)$$

$$\theta_a = 4.04^\circ$$

12. Outer cone angle, $\delta_a = \delta + \theta_a = 45^\circ + 4.04^\circ = \mathbf{49.04^\circ}$

13. Root cone angle, $\delta_f = \delta - \theta_f = 45^\circ - 5.05^\circ = \mathbf{39.95^\circ}$

14. Outside diameter, $d_a = d + 2h_a \cos \delta$

$$d_a = 30 + 2 \times 1.5 \cos 45^\circ$$

$$d_a = \mathbf{32.1 \text{ mm}}$$

15. Pitch apex to crown, $X = R_e \cos \delta - h_a \sin \delta$

$$X = 21.21 \cos 45^\circ - 1.5 \sin 45^\circ$$

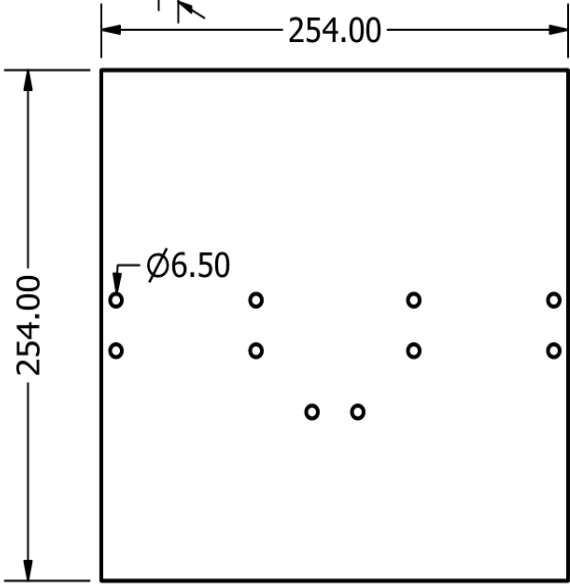
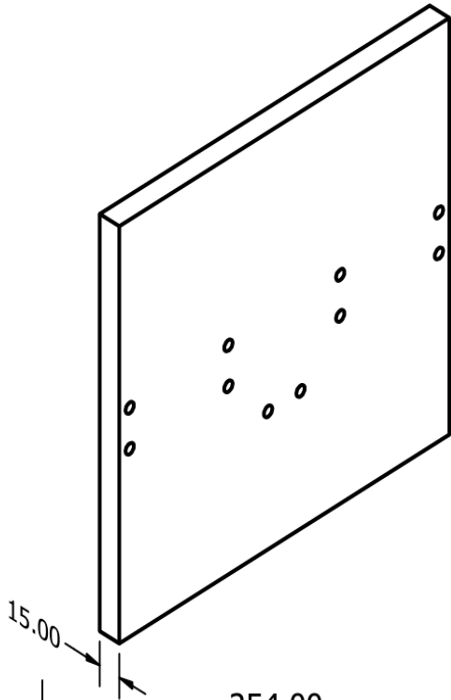
$$X = \mathbf{14 \text{ mm}}$$

Project Parts List

Below is the list of parts with details in a five-column format: Part Name, Quantity, Material Used, Manufactured/Bought, and if manufactured, the machining operations required.

Part Name	Quantity	Material Used	Manufactured/Bought	Machining Operations
Bevel Gear	3	Mild Steel	Manufactured	Lathe, Milling, Taper Turning, Facing, Drilling
Rubber Wheels	4	Rubber	Bought	-
L-Support	6	Mild Steel	Manufactured	Cutting, Filing, Drilling
DC Motor	4	Metal and Plastic	Given	-
Base Plate	1	Mild Steel	Manufactured	Cutting, Drilling
Legs	4	Mild Steel	Manufactured	Cutting, Filing, Drilling
Shaft	3	Mild Steel	Manufactured	Cutting, Turning, Threading
Motor Driver	1	Electronic Components	Bought	-
Arduino UNO Board	1	Electronic Components	Given	-
Screw	11	Steel	Given	-
Battery (DC)	1	Chemical and Metal Components	Given	-

PARTS LIST		
ITEM	QTY	PART NUMBER
4	1	Base Plate

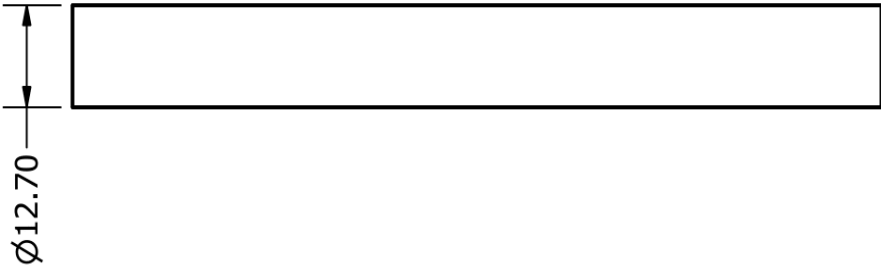
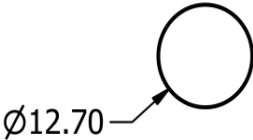
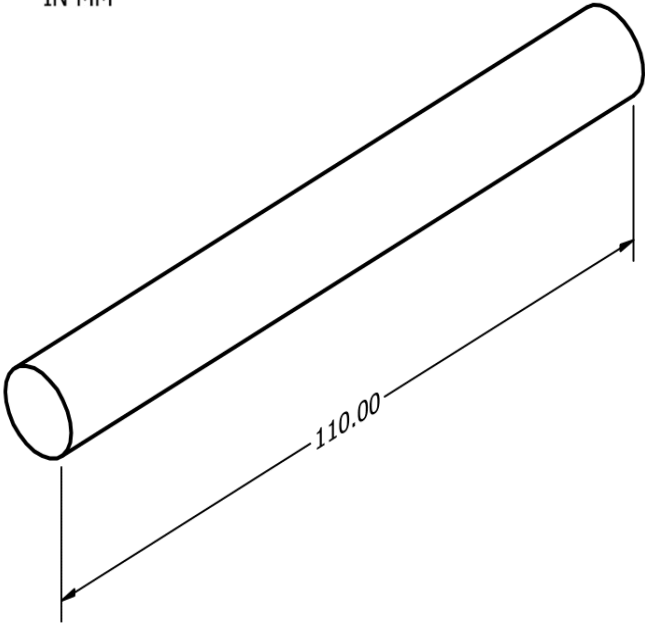
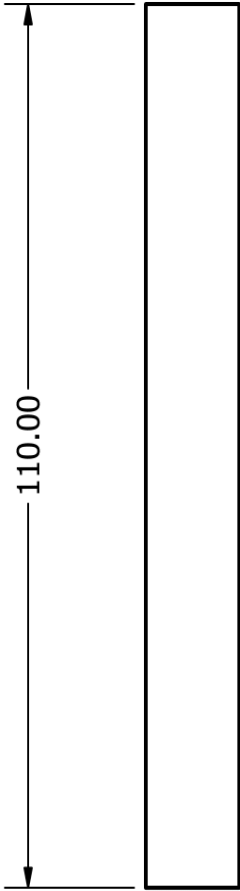


SCALE 1:4
ALL
DIMENSIONS
IN MM

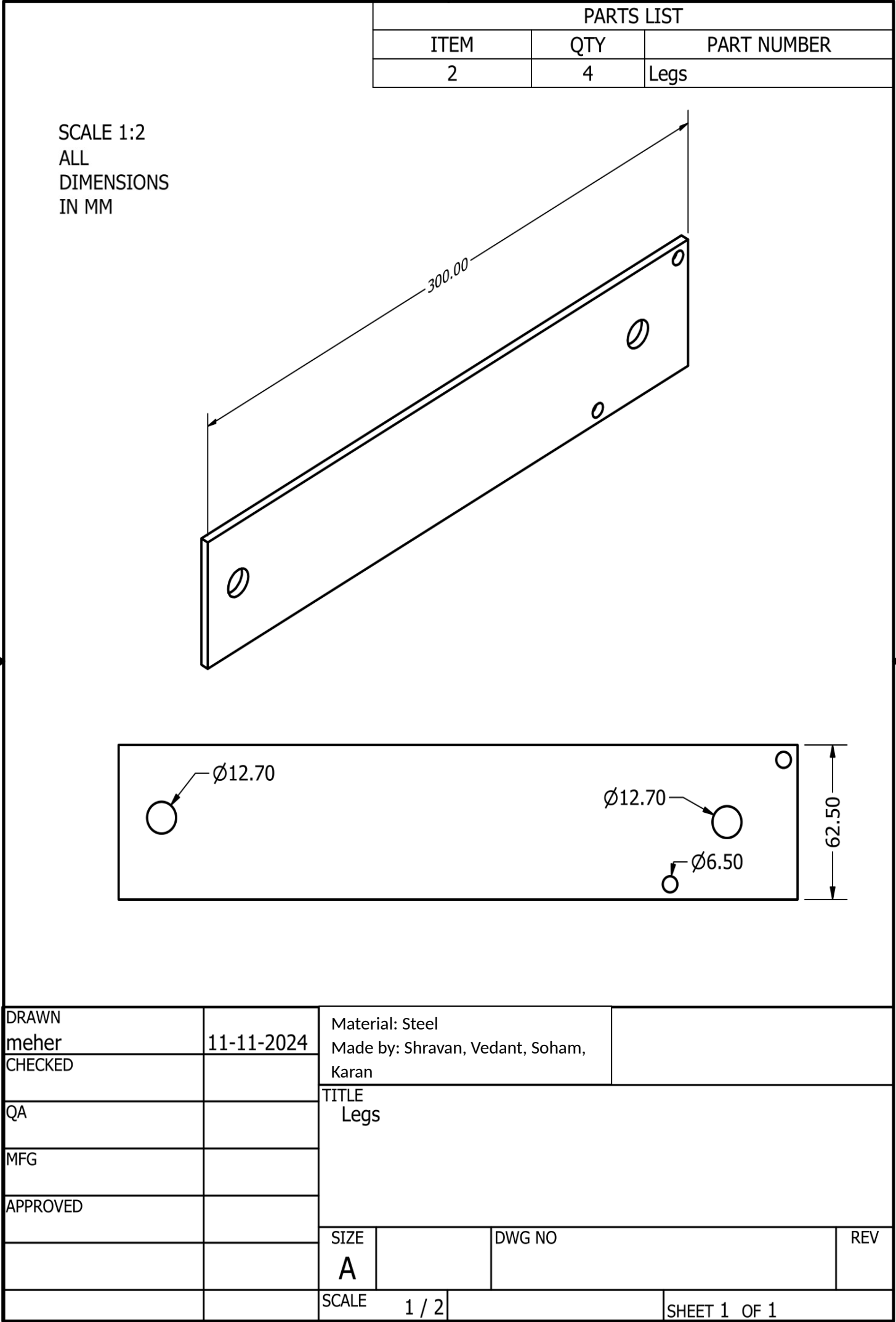
DRAWN meher	11-11-2024	Material: Steel			
CHECKED		Made by: Meher, Keshav, Karan			
QA		TITLE Base Plate			
MFG					
APPROVED					
		SIZE A		DWG NO	REV
		SCALE	1 / 4		SHEET 1 OF 1

PARTS LIST		
ITEM	QTY	PART NUMBER
5	1	Shaft

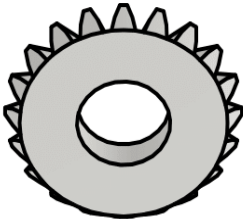
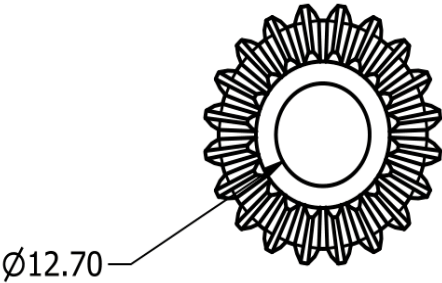
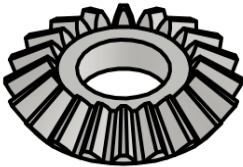
SCALE 1:1
ALL
DIMENSIONS
IN MM



DRAWN meher	11-11-2024	Material: Steel	
		Made by: Keshav, Shravan, Ananya, Ronit	
CHECKED		TITLE Shaft	
QA			
MFG			
APPROVED			
		SIZE A	DWG NO
		SCALE 1 : 1	SHEET 1 OF 1



PARTS LIST		
ITEM	QTY	PART NUMBER
1	3	Bevel Gear



SCALE 1:1
ALL
DIMENSIONS
IN MM

DRAWN	11-11-2024	Material: Steel Made by: Ronit, Ananya, Akul			
meher					
CHECKED		TITLE Bevel Gear Module : 1.5 No. of teeth : 20 Pitch Diameter : 30 mm			
QA					
MFG					
APPROVED					
		SIZE		DWG NO	REV
		A		Bevel_Gear[1]	
		SCALE	1 : 1		SHEET 1 OF 1

Material List_____

S.No	PART NAME	MATERIAL USED	DESCRIPTION	COST OF MATERIA
1.	BASE PLATE x1	MILD STEEL	Dimension(200x200)m m x Thickness 6mm	₹188.4
2.	L-Support x5	MILD STEEL	Dimension(53x50x10)m m Thickness 3mm	₹104
3.	Legs x 4	MILD STEEL	Length 300 mm	₹94
4.	Shafts Long x 1	MILD STEEL	Length 150 mm	₹19.75
5.	Shaft Mid x 2	MILD STEEL	Length 110 mm	₹40.9 x2 =₹81.8
6.	Bevel Gear1 x 2	MILD STEEL	Diameter 30mm Lenath 60mm	₹333 x2 =₹666
7.	Bevel Gear2 x 1	MILD STEEL	Diameter 30mm Lenath 50mm	₹277.94
8.	Wheels x 4	N/A	Diameter 100 mm Width 40mm	₹330
9.	DC Motors x 4	N/A	30 RPM	₹200 x 4 =₹800
10.	Battery x 1	N/A	12 V SMPS	₹350
	TOTAL1			₹2911.89

COST CALCULATION

1.Base Plate:-

Dimension =(200 x 200 x6)mm

Weight = $240 \times 7.85 = 1.884$ kg

Cost = $1.884 \times 100 = \text{Rs.}188.4$

2.L-Support :-

Dimension =(53 x 53 x10)mm

Weight = $26.5 \times 7.85 = 0.208 \times 5 = 1.04$ kg

Cost = $1.04 \times 100 = \text{Rs.}104$

3.Legs

Dimension :- (300 x 10x10)mm
= 60/50

Weight :- $235.5 \text{ grams} \times 4 = 0.94$ kg

Cost = $0.94 \times 100 = \text{Rs.}94$

4.Shaft 1

Dimension = Dia. = 12.7 mm and Length = 150mm

Weight = $19.0247 \times 7.85 = 199.75$ grams = 0.19 kg

Cost = $0.1975 \times 100 = \text{Rs.}19.75$

5.Shaft 2

Dimension= Dia.12.7 mm and Length = 110mm

Weight = $23,967.42 \times 7.85 = 409.65$ grams=0.409 kg

Cost = $0.409 \times 100 = \text{Rs.}40.9$

6. Bevel Gear 1 and 2

Dimension = Dia. = 30 mm; Radius = 15mm Length

Weight = $42,411 \times 7.85 = 0.333$ kg and 0.277 kg

Cost1 = $0.333 \times 100 = \text{Rs.}333$

Cost2 = $0.27794 \times 100 = \text{Rs.}277.94$

MACHINERY COST

MACHINE NAME	TIME SPENT	COST
DRILLING	6 HOURS	$6 \times 75 = \text{Rs.}430$
TURNING	3.5 HOURS	$3.5 \times 150 = \text{Rs.}525$
MILLING	8 HOURS	$8 \times 250 = \text{Rs.}2000$
TOTAL2		RS.2955

$$\begin{aligned}\text{TOTAL COST OF PROJECT} &= \text{Rs.}2955 + \\ &\text{Rs.}2911.89 = \\ &\text{Rs.}5866.89\end{aligned}$$